

Power BI Project — Data Modeler

Data Model Summary Report

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1. Project Overview

Project Name	Data Modeler
Student Name	Ravindra Kumar
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Tool Used	Microsoft Power BI Desktop
File	Data_Modeler.pbix
Report Page	Page 1
Total Tables	6 (2 Fact + 4 Dimension)
Total Records	1,680 rows across all tables
Date Range	FY2022 - FY2023 (2 Fiscal Years)
Schema Type	Star Schema
Total Visuals	3 Matrix (Pivot) Tables

This project demonstrates data modelling in Power BI by building relationships between dimension and fact tables, defining data categories and hierarchies, configuring relationship cardinality and cross-filter directions, and verifying the model using Matrix visuals.

2. Tables in the Data Model

Table Name	Type	Rows	Key Column	Columns in Dataset
Sales_Fact	Fact Table	500	SalesID	SalesID, CustomerID, ProductID, RegionID, DateKey, Quantity, Revenue, Discount
Returns_Fact	Fact Table	100	ReturnID	ReturnID, SalesID, ReturnDateKey, Reason
Customer_Dim	Dimension Table	200	CustomerID	CustomerID, FullName, Age, Gender, Segment

Product_Dim	Dimension Table	100	ProductID	ProductID, ProductName, Category, Subcategory, Brand
Region_Dim	Dimension Table	50	RegionID	RegionID, Country, State, City
Date_Dim	Dimension Table	730	DateKey	DateKey, Date, Month, Quarter, Year, Fiscal Year

3. Relationships & Cardinality

The following relationships were configured in the Model View. All primary relationships use One-to-Many (1:*) cardinality with Single cross-filter direction (Dimension → Fact), following Star Schema best practices.

From Table (Dimension)	Key Column	To Table (Fact)	Cardinality	Cross-Filter	Status
Customer Data	CustomerID	Sales Data	1 : *	Single	Active
Product Data	ProductID	Sales Data	1 : *	Single	Active
Region Data	RegionID	Sales Data	1 : *	Single	Active
Date Data	Date	Sales Data	1 : *	Single	Active
Date Data	Date	Return Data	1 : *	Single	Inactive (dotted)
Return Data	ReturnID	Sales Data	* : *	Single	Reference

Inactive Relationship Note: The Date Data → Return Data relationship on ReturnDate is kept inactive to avoid filter ambiguity. It can be activated using USERELATIONSHIP() in DAX, or by manually toggling in Edit Relationship dialog.

4. Data Categories & Formats

Table	Column	Data Type	Data Category	Format / Values
Region_Dim	Country	Text	Country/Region	USA / India / Germany
Region_Dim	State	Text	State or Province	CA, DL, MH, NY, BW, TX
Region_Dim	City	Text	City	City 1 to City 50
Sales_Fact	Revenue	Decimal	Uncategorized	Currency — Range: 103 to 996
Sales_Fact	Quantity	Whole Number	Uncategorized	Whole Number — Range: 1 to 9
Sales_Fact	Discount	Whole Number	Uncategorized	0%, 5%, 10%, 15%, 20%
Date_Dim	Date	Date	Uncategorized	DD/MM/YYYY
Date_Dim	Fiscal Year	Text	Uncategorized	FY2022 / FY2023
Product_Dim	Category	Text	Uncategorized	Clothing / Electronics / Furniture

Customer_Dim	Segment	Text	Uncategorized	Gold / Silver / Platinum
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5. Hierarchies Built

Hierarchy Name	Levels (Top to Bottom)	Table
Date_Dim Hierarchy	Year > Quarter > Month > Date	Date Data
Region_Dim Hierarchy	Country > State > City	Region Data
Product_Dim Hierarchy	Category > Subcategory > ProductName	Product Data

Hierarchies enable Drill Down functionality in visuals. Created by right-clicking the top-level column and selecting 'Create Hierarchy', then adding child columns via 'Add to Hierarchy'.

6. Matrix Table Verification (Report Page 1)

Three Matrix (Pivot Table) visuals were used to verify the data model. All visuals are on Page 1 of the report.

Visual 1: Sales grouped by Product Category and Region

Rows	Product Data[Category]
Columns	Region Data[Country]
Values	Sum of Sales Data[Revenue]
Tables Used	Product Data, Sales Data, Region Data

Visual 2: Return Reasons by Fiscal Year

Rows	Return Data[Reason]
Columns	Date Data[Fiscal Year]
Values	Count of Return Data[ReturnID]
Tables Used	Return Data, Date Data

Visual 3: Revenue by Customer Segment

Rows	Customer Data[Segment]
Columns	(Empty — single column layout)
Values	Sum of Sales Data[Revenue]
Tables Used	Customer Data, Sales Data

7. Challenges Faced & How They Were Solved

#	Challenge	Solution Applied
1	Inactive relationship between Date & Return Data causing filter ambiguity	Kept one relationship active (OrderDate); ReturnDate relationship marked inactive (dotted line). USERELATIONSHIP() can activate it in DAX measures.
2	Bidirectional filters causing circular filter paths	Set all relationships to Single cross-filter direction; Bidirectional enabled only where Many-to-Many required it.
3	Many-to-Many ambiguity risk between Return Data and Sales Data	Identified and resolved by reviewing cardinality; adjusted join key or used bridge table approach.
4	Data Categories not set for geographic columns	Manually set Country/Region, State or Province, City categories via Column Tools tab for correct map rendering.
5	Hierarchies missing for drill-down in visuals	Built Date, Region, and Product hierarchies manually in Model View using right-click > Create Hierarchy workflow.

8. Data Model Summary

Schema Type	Star Schema — 2 Fact tables at center, 4 Dimension tables around
Total Tables	6 (2 Fact + 4 Dimension)
Total Records	1,680 rows across all 6 tables
Date Range	FY2022 - FY2023 (730 days)
Total Relationships	6 (5 Active, 1 Inactive)
Hierarchies Built	3 (Date_Dim, Region_Dim, Product_Dim)
Visuals on Report	3 Matrix Tables
Cross-Filter Default	Single (Dimension to Fact)
Data Categories Set	Country/Region, State or Province, City — Region_Dim table
Key Business Values	Revenue: 103-996 Discount: 0-20% Segments: Gold/Silver/Platinum Categories: Clothing/Electronics/Furniture